

Cover letter — Scientific Reports

Dear Editors,

We submit our manuscript, “**Sensitivity of AI-Generated Credit Ratings to Elicitation Choices,**” for consideration as an Article in *Scientific Reports*.

Large language models are increasingly used as credit, risk, and trading analysts, on the tacit assumption that a model holds a stable latent assessment that the prompt merely reads out. We test that assumption directly with a **pre-registered specification-curve design**: a fixed 90-item firm-profile battery with an objective Altman Z' benchmark, crossed with a factorial grid of defensible elicitation choices (provider, model version, temperature, prompt paraphrase, output format, few-shot exemplars, and answer presentation), replicated across seeds and three model families (12,960 elicitations). We find that although firm fundamentals explain most rating variation, the machine credit rating retains economically meaningful sensitivity to defensible elicitation choices and to residual run-to-run variation, more than to any single tunable design choice; that the fragility survives honored determinism and cannot be escaped by switching or pooling vendors; and that the opinion holds no stable rating resolution, crossing the investment-grade/high-yield line on 27% of matched elicitation pairs. We translate this into the units a risk function acts on—portfolio turnover, credit spread, and a post-hoc Basel capital recast—and, to answer the “use real data” objection, we first **demonstrate that naive real-firm LLM evaluation can be contaminated by memorization** and then show, under a pre-registered fingerprinting decontamination gate, that **substantial specification instability is also observed on real anonymized issuers** benchmarked to disclosed agency ratings.

We believe the work fits *Scientific Reports* for three reasons. (i) **Broad relevance**: it concerns the reliability of AI decision systems in a high-stakes deployment, of interest across machine learning, finance, and metascience. (ii) **Methodological rigor**: the design is pre-registered, the analyses are confirmatory with permutation tests and cluster-bootstrap intervals, and the decontamination gate is, to our knowledge, the first direct evidence that anonymized real-firm LLM evaluation can be memorization-contaminated even without names in the prompt. (iii) **Full reproducibility**: the entire 12,960-response corpus, code, and manifest hashes are released; every reported number regenerates offline and byte-for-byte from the frozen data.

The manuscript is original, not under consideration elsewhere, and all authors approve the submission. We declare no competing interests. The corresponding author is Robin Chawla (robin.chawla.cse14@iitbhu.ac.in).

Referees to exclude. None.

Prior discussions. We have not had any prior discussions with a Scientific Reports Editorial Board Member about the work described in this manuscript.

Thank you for your consideration.

Sincerely, Samir Chincholikar and Robin Chawla Independent researchers, New York, USA